KARTAVYA GARHWAL

B.Tech in Computer Science | Aspiring Full-Stack & AI/ML Developer

Jaipur, Rajasthan, India | JK Lakshmipat University (2024–2028) | Third Year
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PROFESSIONAL SUMMARY

Computer Science Engineering undergraduate at JK Lakshmipat University (JKLU), Jaipur, with strong academic footing and practical experience in MERN stack development, Python systems, Machine Learning, and Cybersecurity. Passionate about building performant web applications, data-driven forecasting models, and Retrieval-Augmented Generation (RAG) architectures.

EDUCATION

Bachelor of Technology (B.Tech) in Computer Science — 2024 – 2028
JK Lakshmipat University (JKLU), Jaipur, Rajasthan
Current Academic Year: Third Year | Relevant Coursework: Data Structures & Algorithms, Database Management Systems, Full-Stack Web Development, Machine Learning Fundamentals, Computer Networks, Operating Systems.

TECHNICAL SKILLS

Programming Languages: C, Python, JavaScript (ES6+), SQL
Frontend Development: React.js, HTML5, CSS3/Modern CSS, Vite, Responsive Web Design
Backend Development: Node.js, Express.js, Django, RESTful API Architecture
Databases: MongoDB, Mongoose ODM, Relational SQL
Data Science & AI/ML: pandas, NumPy, scikit-learn, Retrieval-Augmented Generation (GraphRAG), Anomaly Detection
Tools & DevOps: Git, GitHub, npm, VS Code, Postman, Linux Basics

KEY ACADEMIC & TECHNICAL PROJECTS

End-to-End Device Monitoring Solution *[Python, Node.js, Express, MongoDB, WebSockets, Cybersecurity]*
- Engineered a centralized cybersecurity monitoring system capturing device metrics, active processes, and network socket traffic.
- Implemented anomaly detection heuristics and real-time security event alerts integrated into an intuitive monitoring dashboard.
- Secured communication pipelines and logged host security status for operational auditing.

GraphRAG Knowledge Retrieval System *[Python, Knowledge Graphs, Vector Search, LLM RAG, NLP]*
- Designed a hybrid QA system fusing unstructured document retrieval with knowledge-graph entity relationships.
- Enabled multi-hop reasoning over complex technical corpora, connecting fragmented facts across multiple documents.
- Optimized context synthesis to reduce hallucination and enhance response explainability.

Air Quality Index (AQI) Analysis & Prediction *[Python, pandas, NumPy, scikit-learn, Data Visualization]*
- Constructed an exploratory data pipeline analyzing particulate matter (PM2.5, PM10), CO, and NO2 pollution indicators.
- Evaluated correlation matrices and trained baseline regression models to accurately forecast environmental AQI levels.

Student Academic Recovery Prediction *[Python, pandas, scikit-learn, Feature Engineering]*
- Conducted exploratory data analysis on student academic progression patterns and assessment indicators.
- Built predictive baseline models identifying early signals for academic intervention and student performance recovery.

CERTIFICATIONS, ACHIEVEMENTS & INTERESTS

- Academic Projects: Actively pursuing practical engineering implementations in full-stack web and intelligent systems.
- Professional Focus: Cybersecurity architectures, scalable backend services, and Retrieval-Augmented Generation workflows.
- *[Note: Verified certifications from NPTEL/Coursera/platforms can be added in the centralized portfolio configuration]*